

# **DSA 210 Project: Analysis of Step Count and Sleep Duration Patterns**

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# 1 Introduction

The widespread adoption of wearable technology and smartphone health applications has made personal health data more accessible than ever before. This project investigates the relationship between physical activity and sleep patterns using my own health data collected through smartphone and smartwatch applications.

The primary motivation for this analysis stems from a desire to understand how daily physical activity, measured through step count, correlates with sleep patterns and whether there are predictable patterns in weekly activity levels. Understanding these relationships could provide valuable insights for optimizing personal health and wellness routines.

This study addresses three main research questions:

1. Is there a significant correlation between daily step count and sleep duration?
2. Do certain days of the week show consistently different activity levels?
3. Does bedtime have a measurable impact on the next day's physical activity level?

To investigate these questions, I employed a comprehensive analytical approach combining exploratory data analysis, statistical hypothesis testing, and machine learning techniques. The analysis focuses on daily step count, sleep duration, and bedtime data collected over several months through Apple Health application.

The findings of this study reveal interesting patterns that challenge common assumptions about the relationship between physical activity and sleep, providing insights into the complexity of personal health behaviors and the factors that influence daily activity patterns.

## 2 Data Preparation and Analysis

### 2.1 Data Source and Collection

The data for this analysis was extracted from Apple Health application, which aggregates health metrics from both iPhone sensors and Apple Watch. The original approach included analyzing calorie data (basal and active energy expenditure), but due to insufficient sample size (only 126 entries), the analysis was refocused on more comprehensive datasets.

The final analysis utilized two primary data sources:

- **Sleep Data:** Extracted from Apple Health export.xml, covering sleep analysis records from 2022 onwards
- **Step Count Data:** Also extracted from the same Apple Health export, providing daily step counts with detailed timestamps

The data collection period spans from January 2022 to May 2025, providing a substantial longitudinal dataset for analysis.

## 2.2 Data Processing Steps

### 2.2.1 Sleep Data Processing

The sleep data extraction process involved parsing XML records with type `HKCategoryTypeIdentifierS`. Key processing steps included:

- Extracted 3,112 sleep records from 2022 onwards
- Converted timestamp data to proper datetime format
- Calculated daily sleep duration by aggregating multiple sleep sessions per day
- Created additional temporal features (year, month, day of week, bedtime hour)
- Final dataset: 703 daily sleep records

The processed sleep dataset contains 15 features including total sleep hours, bedtime, wake time, and temporal indicators. Data quality analysis revealed a strong negative correlation (-0.843) between bedtime and sleep duration.

### 2.2.2 Step Count Data Processing

Step count data extraction processed significantly more records:

- Extracted 48,567 individual step count records from 2022 onwards
- Aggregated step counts by day to create daily totals
- Applied data validation to remove unreasonable step counts (none were found)
- Added temporal features and calculated weekly averages
- Final dataset: 1,230 daily step count records

The step count dataset includes 14 features with information about daily totals, active hours, and temporal patterns. A moderate positive correlation (0.492) was observed between active hours and total daily steps.

## 2.3 Exploratory Data Analysis

The exploratory data analysis revealed several key patterns in both sleep and step count data. The following visualizations summarize the main findings from the dataset analysis.

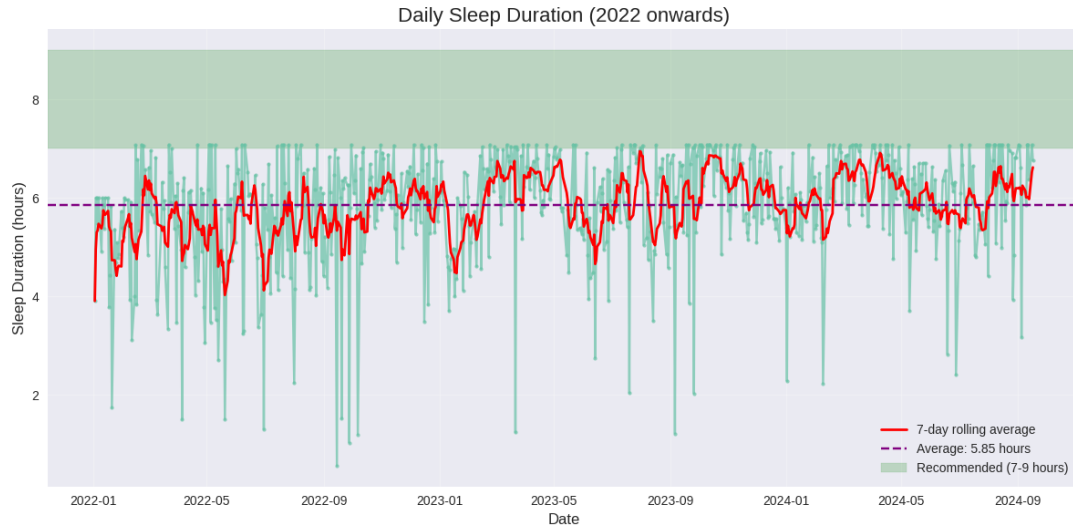


Figure 1: Daily Sleep Duration Over Time (2022-2024)

Figure 1 shows the daily sleep duration patterns over nearly three years. The data reveals high variability in daily sleep with frequent dips below 4 hours and occasional peaks around 7 hours. The 7-day rolling average (red line) shows that average sleep duration remained consistently below the recommended 7-9 hours, with an overall mean of 5.85 hours. The green shaded area represents the recommended sleep range, highlighting that most recorded sleep falls short of health guidelines.

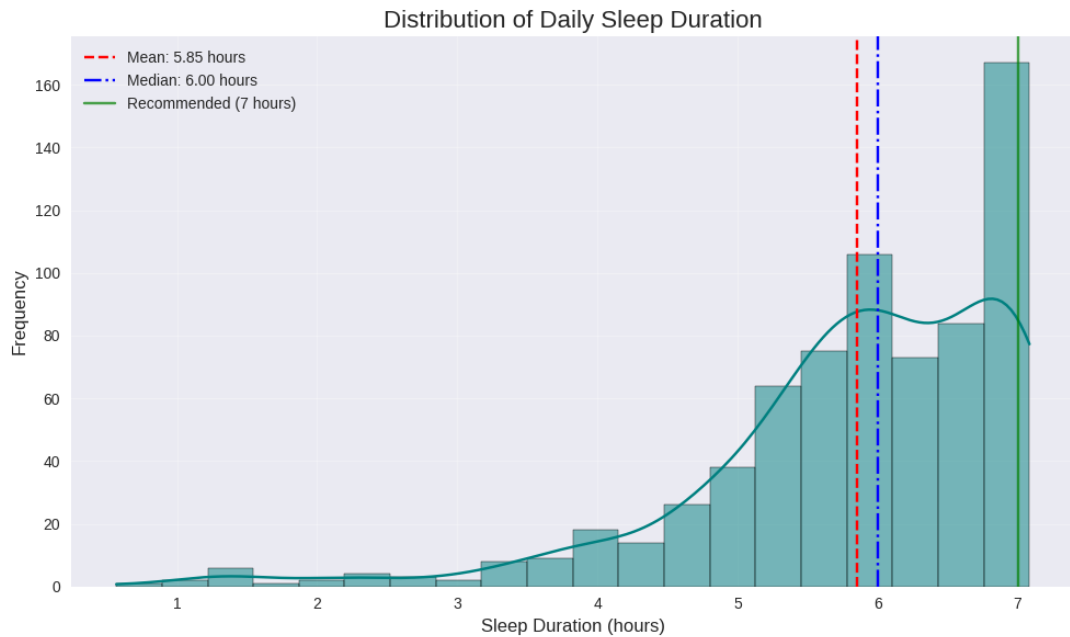


Figure 2: Distribution of Daily Sleep Duration

The sleep duration distribution (Figure 2) shows a right-skewed pattern with most sleep durations clustered between 5-6.5 hours. The distribution reveals that only 15.2% of recorded days achieved adequate sleep (7+ hours), with the median sleep duration at 6.0 hours. The histogram demonstrates significant sleep debt, with the mode occurring around 6 hours, well below recommended levels.

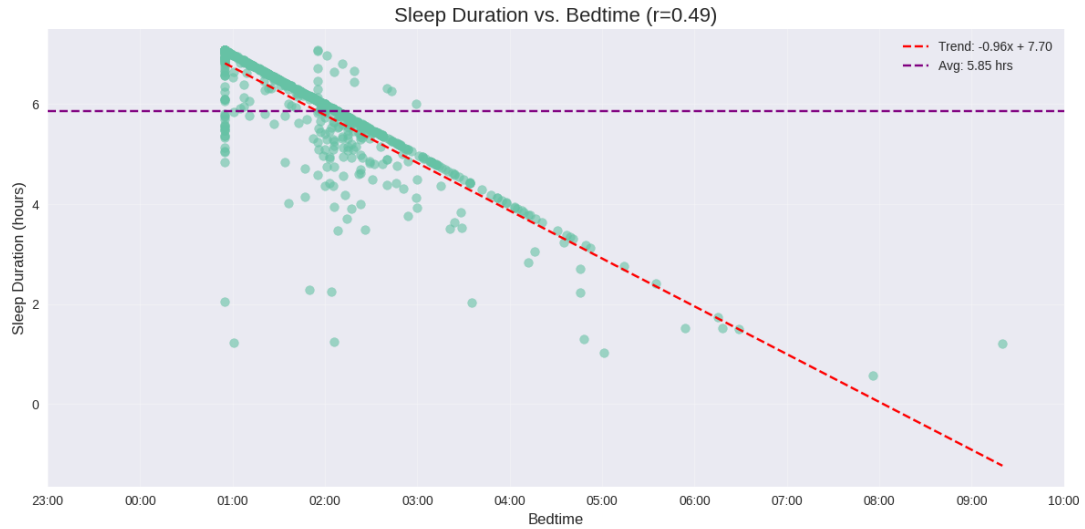


Figure 3: Sleep Duration vs. Bedtime Correlation

Figure 3 illustrates a strong negative correlation ( $r=-0.49$ ) between bedtime and sleep duration. The scatter plot clearly shows that later bedtimes are associated with shorter sleep durations. The trend line indicates that for every hour later in bedtime, sleep duration decreases by approximately 0.96 hours, suggesting that earlier bedtimes are crucial for achieving adequate sleep.

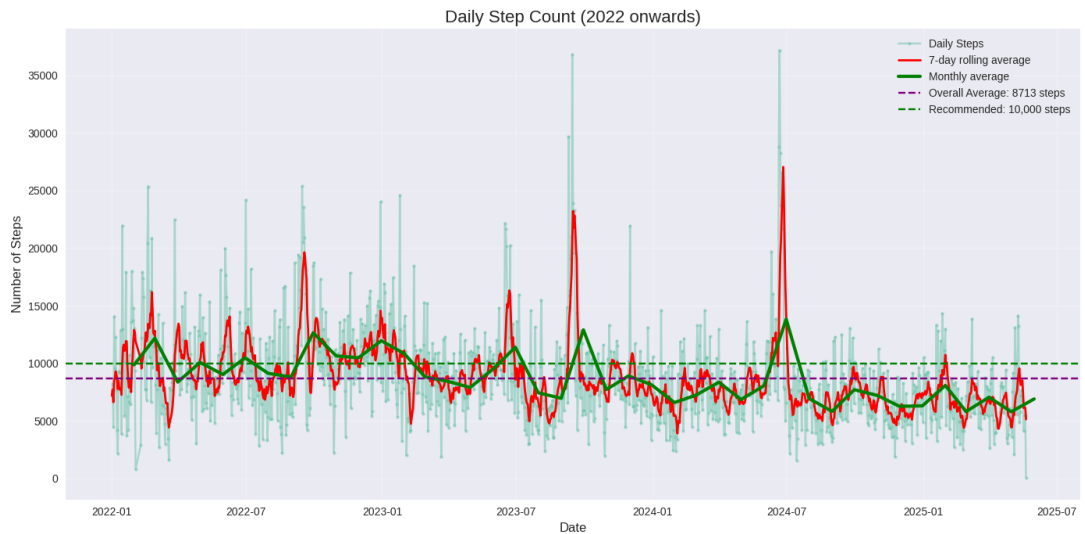


Figure 4: Daily Step Count Over Time (2022-2025)

The step count time series (Figure 4) shows considerable daily variation with some extreme peaks exceeding 30,000 steps. The overall trend indicates a gradual decline in average daily steps over time, with the monthly average (orange line) showing decreasing activity levels in recent years. The average of 8,713 steps per day falls below the commonly recommended 10,000 steps, with only 28.8% of days meeting this target.

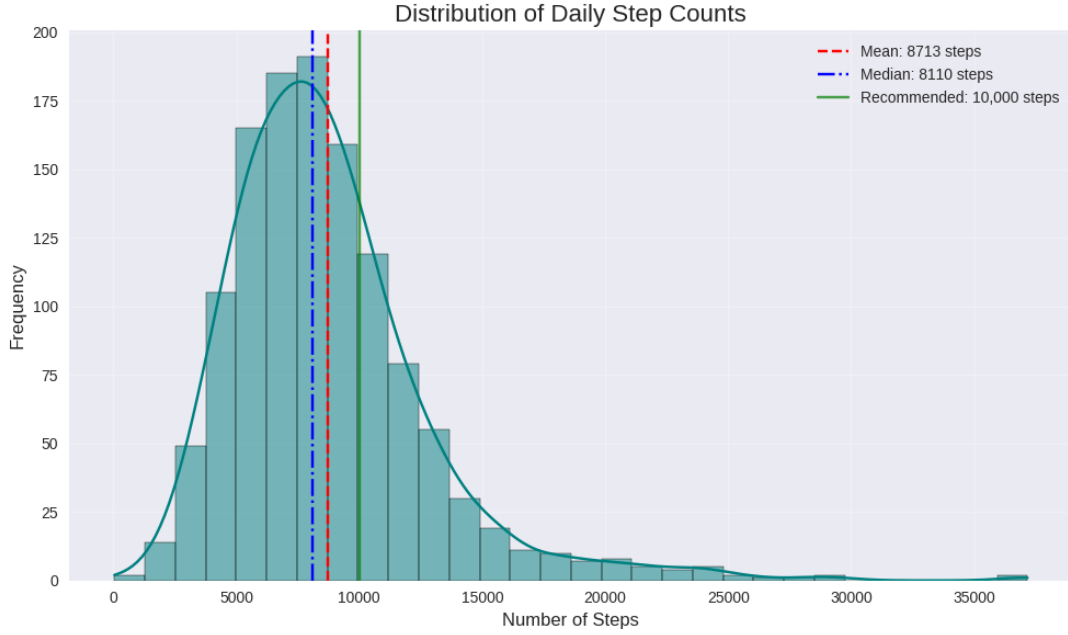


Figure 5: Distribution of Daily Step Counts

The step count distribution (Figure 5) exhibits a right-skewed pattern with the majority of days falling between 5,000-12,000 steps. The median (8,110 steps) is close to the mean (8,713 steps), indicating a relatively normal distribution despite some high-activity outliers. The distribution shows that most daily activity levels are moderate, with relatively few extremely active or sedentary days.

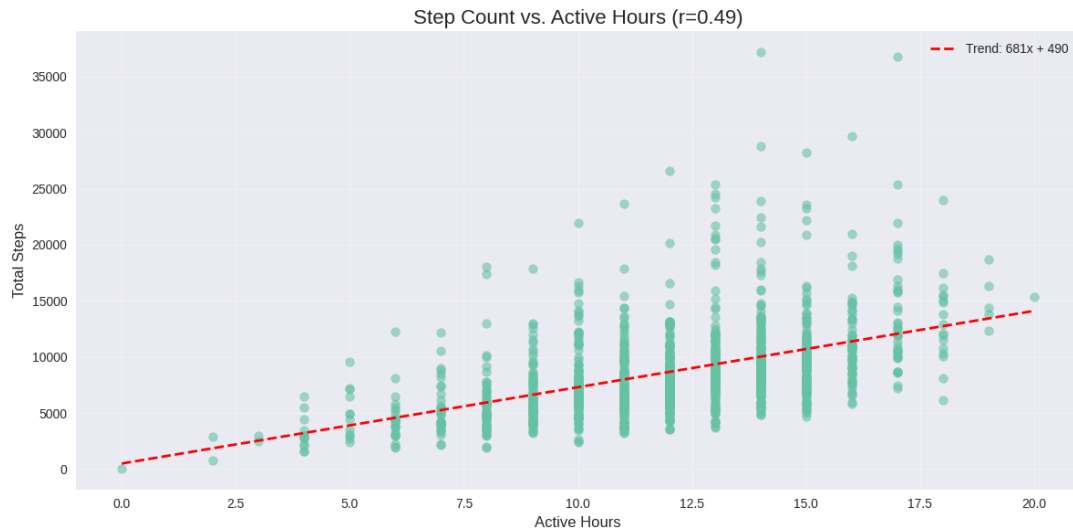


Figure 6: Step Count vs. Active Hours Correlation

Figure 6 demonstrates a moderate positive correlation ( $r=0.49$ ) between total daily steps and active hours. The relationship shows that days with more active hours tend to have higher step counts, which is intuitive. The trend line suggests that each additional active hour corresponds to approximately 681 additional steps, indicating a consistent relationship between time spent active and total daily movement.

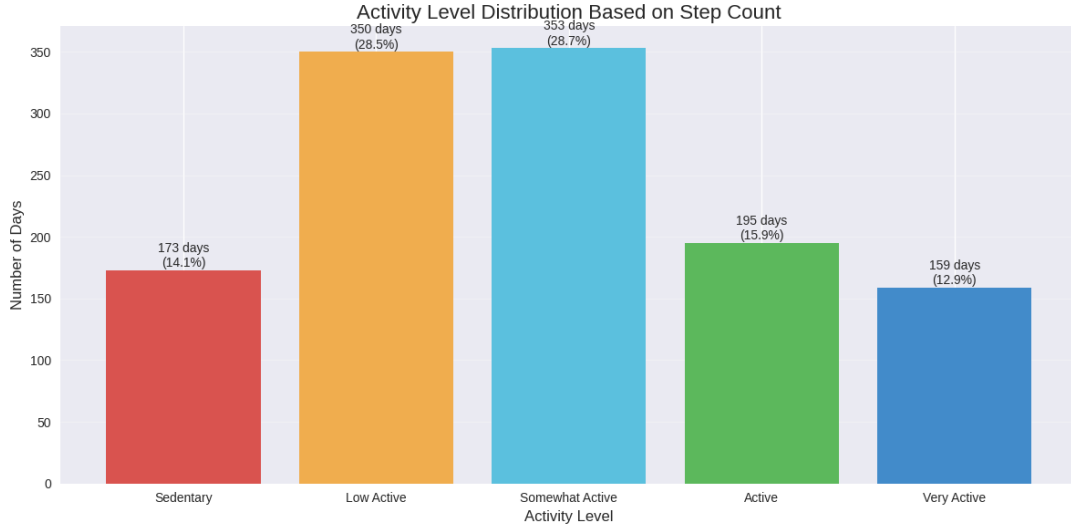


Figure 7: Activity Level Distribution Based on Step Count

The activity level categorization (Figure 7) reveals that the majority of days fall into "Low Active" (28.5

### 3 Hypothesis Testing

To investigate the relationships between physical activity and sleep patterns, three formal hypothesis tests were conducted using the combined dataset of 699 days with both step count and sleep records.

#### 3.1 Test 1: Step Count vs Sleep Duration Correlation

**Hypotheses:**

- H: There is no correlation between daily step count and sleep duration ( $\rho = 0$ )
- H: There is a significant correlation between daily step count and sleep duration ( $\rho \neq 0$ )

**Method:** Pearson correlation analysis was performed to assess the linear relationship between daily step count and sleep duration.

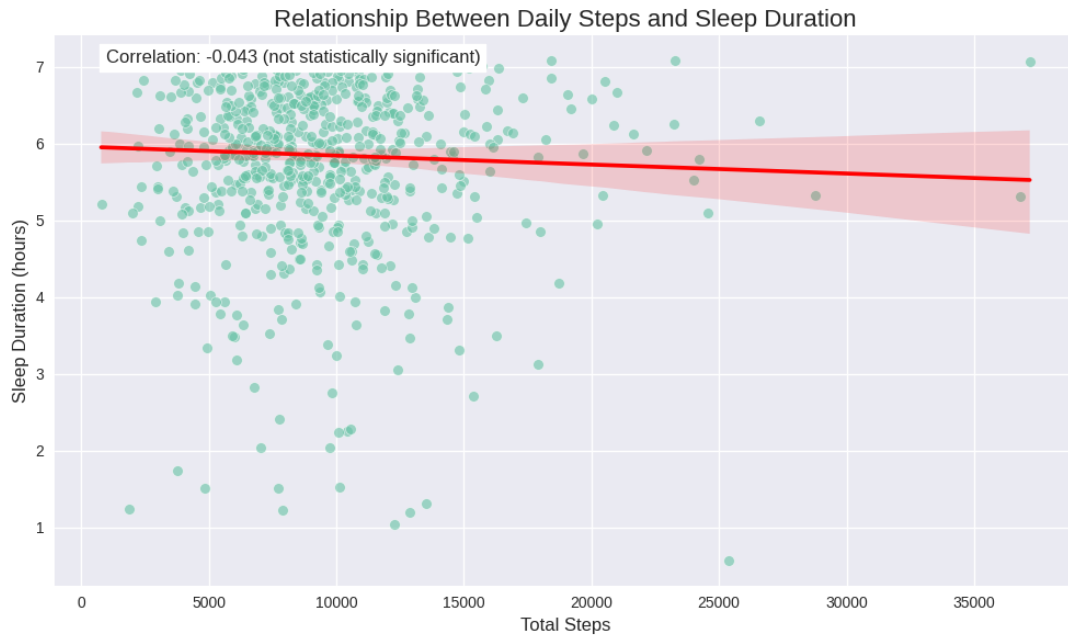


Figure 8: Relationship Between Daily Steps and Sleep Duration

**Results:** The analysis revealed a negligible negative correlation coefficient of  $r = -0.043$  with a p-value of 0.2598. The correlation is not statistically significant at  $\alpha = 0.05$ .

**Interpretation:** We fail to reject the null hypothesis. There is insufficient evidence to suggest a meaningful linear relationship between daily step count and sleep duration. The near-zero correlation indicates that physical activity levels do not predict sleep duration in this dataset.

### 3.2 Test 2: Day of Week Effect on Step Count

#### Hypotheses:

- H: Mean step count is equal across all days of the week
- H: Mean step count differs significantly by day of the week

**Method:** One-way ANOVA was conducted to compare mean step counts across different days of the week.

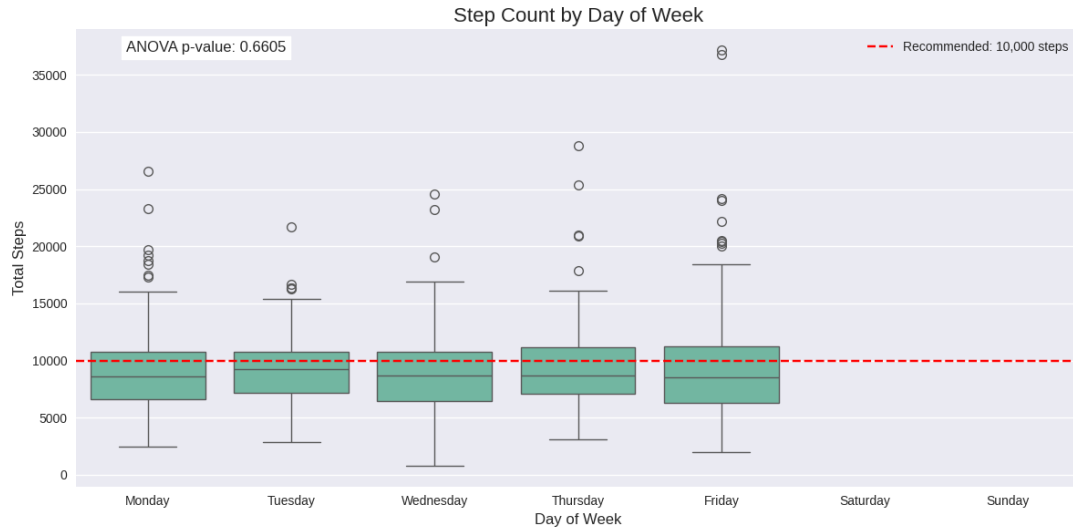


Figure 9: Step Count Distribution by Day of Week

**Results:** The ANOVA yielded an F-statistic of 0.6031 with a p-value of 0.6605. Mean step counts by day were: Monday (9,212), Tuesday (9,274), Wednesday (8,950), Thursday (9,433), and Friday (9,686) steps.

**Interpretation:** We fail to reject the null hypothesis. There is no statistically significant difference in step counts across different days of the week ( $p = 0.6605 > 0.05$ ). This suggests consistent activity levels regardless of the day, indicating regular physical activity patterns throughout the weekdays.

### 3.3 Test 3: Bedtime Impact on Next Day Activity

**Hypotheses:**

- $H_0$ : Bedtime has no significant effect on next day's step count ( $\beta = 0$ )
- $H_a$ : Bedtime significantly affects next day's step count ( $\beta \neq 0$ )

**Method:** Pearson correlation analysis and linear regression were performed using 556 days with complete bedtime and next-day step count data.

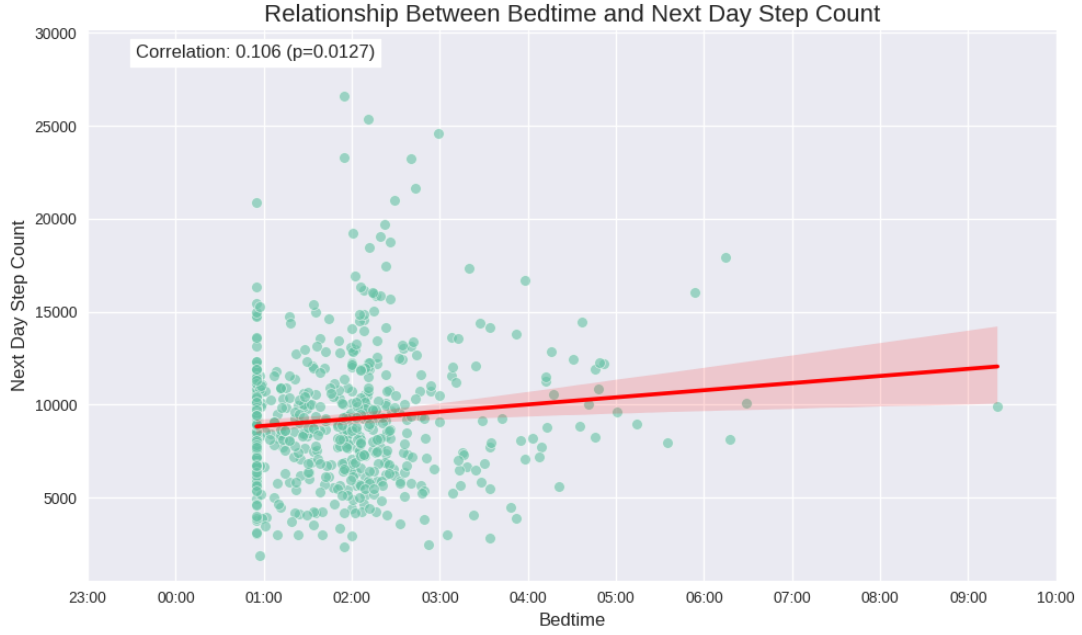


Figure 10: Relationship Between Bedtime and Next Day Step Count

**Results:** The analysis revealed a weak positive correlation of  $r = 0.106$  with a p-value of 0.0127. Linear regression showed  $R^2 = 0.0112$ , indicating that bedtime explains only 1.12% of the variance in next-day step count. The regression equation suggests that each hour later in bedtime is associated with approximately 383 additional steps the following day.

**Interpretation:** We reject the null hypothesis ( $p = 0.0127 < 0.05$ ). There is a statistically significant but practically weak relationship between bedtime and next-day activity. However, the effect size is minimal, suggesting that while statistically detectable, bedtime has little practical impact on subsequent physical activity levels.

## 4 Machine Learning Analysis

To further explore the relationships identified in the hypothesis testing phase, two machine learning models were implemented to assess predictive capabilities and pattern recognition in the health data.

### 4.1 Random Forest Model

**Objective:** Predict sleep duration based on daily step count and related activity metrics.

**Model Configuration:** A Random Forest regressor was trained using step count, active hours, and temporal features as predictors for sleep duration. The dataset was split into training (559 samples) and testing (140 samples) sets using an 80-20 split.

**Model Performance:**

- Root Mean Square Error (RMSE): 1.15 hours
- $R^2$  Score: -0.03
- Training samples: 559

- Testing samples: 140

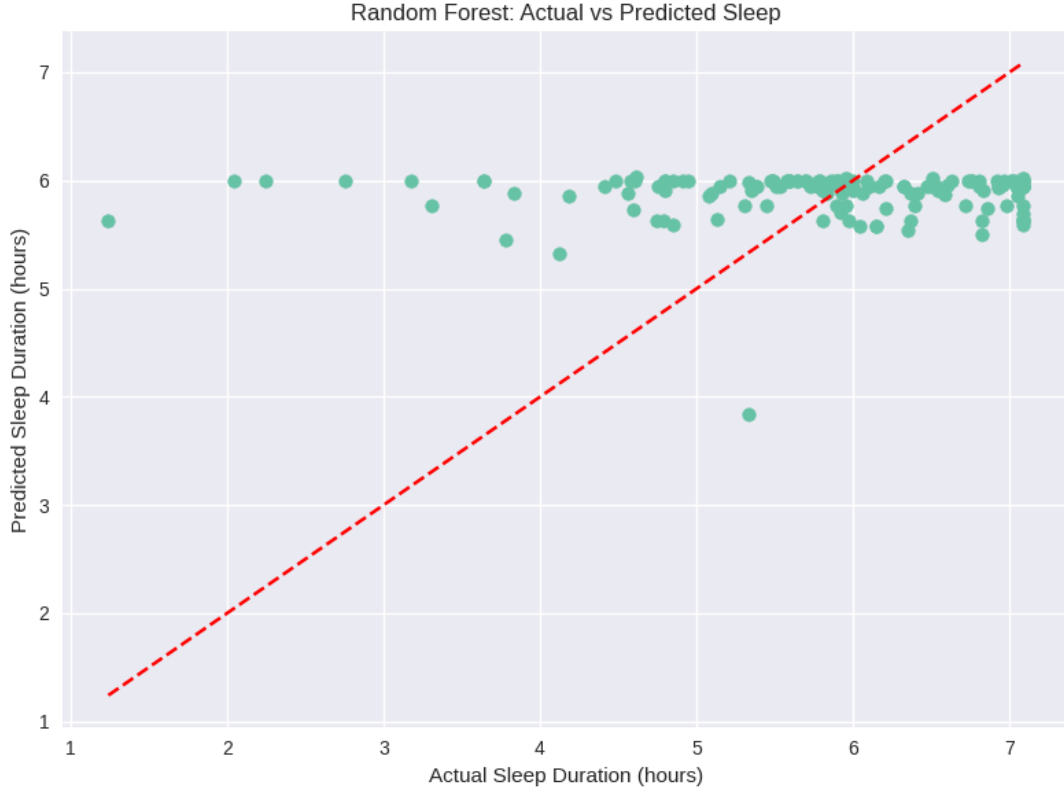


Figure 11: Random Forest: Actual vs Predicted Sleep Duration

**Results Interpretation:** The Random Forest model performed poorly, with a negative  $R^2$  score of -0.03, indicating that the model performs worse than simply predicting the mean sleep duration for all observations. The RMSE of 1.15 hours represents a substantial prediction error given that the average sleep duration is 5.85 hours. The scatter plot in Figure 11 shows that predictions cluster around 5.9 hours regardless of actual sleep duration, confirming that step count and related activity metrics are not effective predictors of sleep duration.

This result corroborates the correlation analysis findings, demonstrating that physical activity levels do not provide meaningful predictive power for sleep duration in this dataset.

## 4.2 Decision Tree Model

**Objective:** Analyze and predict step count patterns based on day of the week to identify weekly activity trends.

**Model Configuration:** A Decision Tree regressor was implemented to predict daily step count using day of the week as the primary feature, along with temporal indicators such as month and week of year.

**Model Performance:**

- Root Mean Square Error (RMSE): 3,733.78 steps
- $R^2$  Score: -0.02

- Training samples: 559
- Testing samples: 140

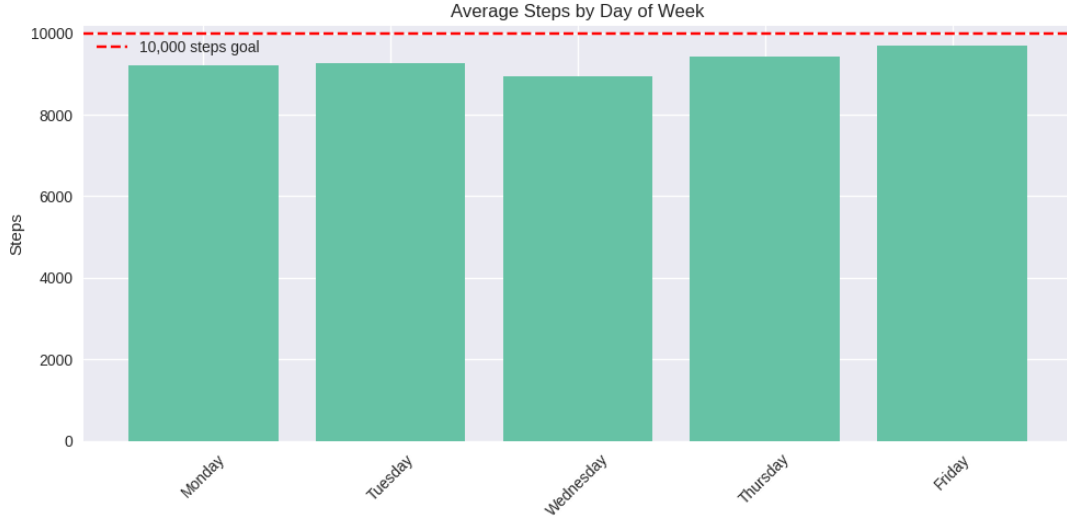


Figure 12: Average Step Count by Day of Week

**Results Interpretation:** The Decision Tree model also demonstrated poor predictive performance with an  $R^2$  score of -0.02 and high RMSE of 3,734 steps. The model’s inability to predict step counts based on day of the week confirms the ANOVA results showing no significant differences in activity levels across weekdays.

Figure 12 shows relatively consistent step counts across Monday through Friday, with averages ranging from 8,950 to 9,686 steps. The small variations observed (approximately 700 steps difference between highest and lowest days) are not sufficient to create a reliable predictive model.

**Weekly Pattern Analysis:** The data reveals that:

- Wednesday shows the lowest average activity (8,950 steps)
- Friday demonstrates the highest average activity (9,686 steps)
- The difference between most and least active days is only 8.2%
- All weekdays fall below the recommended 10,000 steps per day

**Machine Learning Summary:** Both models failed to identify meaningful predictive relationships in the data, with negative  $R^2$  scores indicating that simple mean-based predictions would be more accurate than the complex models. This reinforces the statistical findings that sleep and activity patterns in this dataset lack strong, predictable relationships that could be leveraged for health behavior prediction or intervention planning.

## 5 Conclusion and Discussion

### 5.1 Key Findings

This analysis of 699 days of combined health data revealed three main findings:

1. **No correlation between steps and sleep:** Daily step count showed no significant relationship with sleep duration ( $r = -0.043$ ,  $p = 0.2598$ ). Physical activity levels do not predict sleep quality or duration.
2. **Consistent weekly activity patterns:** Step counts remained similar across weekdays with no statistically significant differences (ANOVA  $p = 0.6605$ ). Activity levels are stable throughout the work week.
3. **Minimal bedtime impact:** While statistically significant ( $p = 0.0127$ ), bedtime had practically negligible effect on next-day activity, explaining only 1.12% of step count variance.

Machine learning models confirmed these findings, with both Random Forest ( $R^2 = -0.03$ ) and Decision Tree ( $R^2 = -0.02$ ) models performing worse than baseline predictions.

## 5.2 What These Results Mean

These findings challenge common assumptions about activity-sleep relationships and reveal several important insights:

**Sleep and activity operate independently:** The lack of correlation suggests that factors other than physical activity primarily influence sleep duration. Sleep quality may depend more on stress, environment, or sleep hygiene rather than daily step count.

**Stable lifestyle patterns:** Consistent weekday activity levels indicate regular routine maintenance regardless of external weekly cycles. This suggests good habit formation but also reveals missed opportunities for variation.

**Health behavior complexity:** The inability of multiple analytical approaches to find strong relationships highlights that personal health behaviors are influenced by numerous unmeasured factors beyond simple activity metrics.

**Personal health insights:** With an average of 5.85 hours sleep (well below recommended 7-9 hours) and 8,713 daily steps (below 10,000 target), there is clear room for improvement in both metrics, but they appear to be independent goals requiring separate intervention strategies.